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United Nations Development Programme

Bosnia and Herzegovina

Initiation Plan

Project Title:	***Localized "Green Deal" - Decarbonization of industry and residential sector in pilot areas***
Expected UNDAF/CP Outcome(s):	**By 2025 people benefit from resilient, inclusive and sustainable growth ensured by the convergence of economic development, and management of environmental and cultural resources**
Expected CPD Output(s):	**Government at all levels, particulary subnational governments, implement climate change adaptation and mitigation measures for resilient and sustainable development.**
Initiation Plan Start/End Dates:	November 01, 2020 to October 31, 2021 (12 months)
Implementing Partner:	UNDP

Brief Description

Bosnia and Herzegovina regularly tops charts in breaching air quality standards with GHG emissions per unit of GDP four times higher than EU average and the highest air pollution in Europe. High levels of air pollution, especially in winter months present a serious threat to human health and sustainable development.

The intervention aims to support authorities to ensure sustainable implementation of energy efficiency and decarbonization measures by providing technical and economic stimulus to GHG emission reduction, air quality improvement, creation of new jobs, budgetary savings/ re-generating resources for investment etc.

This initiation plan focuses on development and implementation of mechanisms for financing energy efficiency and decarbonization measures in residential and SME sectors. Techno-economic analysis will serve as a basis for development of this Mechanism and will provide overview of current situation in these two target sectors, proposed energy efficiency measures, cantonal financial and organizational mechanisms for their implementation and their energy-related, environmental (including air quality), fiscal and job-creation benefits.

By implementing pilot structured measures and building system that is well designed we would be able to achieve sustainable and measurable results that have potential to be replicated elsewhere in the country and scaled up on all government levels in Bosnia and Herzegovina.

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Linkage with SDGs: SDG Goals 1, 7, 9, 11 and 13

Linkage with EU accession agenda: EU Acquis, Chapter 15: Energy, Chapter 27: Environment and climate change

Areas of work aligned with EU Green Deal: a) supplying clean, affordable and secure energy, b) mobilizing industry for a clean and circular economy; c) building and renovating in an energy and resource efficient way; d) reducing air and industrial pollution; e) GHG reduction

Linkage with UNDP Strategic Plan 2018-2021: Accelerating structural transformations for sustainable development, especially through innovative solutions that have multiplier effects across the SDGs.

Contributing Outcome (UNDAF/CPD): Outcome1.By 2025, people benefit from resilient, inclusive and sustainable growth ensured by the convergence of economic development, and management of environment and cultural resources.

Contributing Output (CPD 2021-2025):

Government at all levels, particulary subnational governments, implement climate change adaptation and mitigation measures for resilient and sustainable development.

RELATED STRATEGIC PLAN OUTCOME: Outcome 1: Advance poverty eradication in all its forms and dimensions; Outcome 2: Accelerate structural transformations for sustainable development

Atlas Award: BIH10/00131266

Output: 00124357

Output (with gender marker): GEN2

Total resources required	USD 2,260,000
Total allocated resources:	USD 2,260,000

- Other:
  - UNDP USD 465,000
  - Ongoing projects USD 1,125,000
- Parallel contributions:
  - Government USD 200,000
  - Private/residential s.: USD 470,000

Agreed by UNDP:

Steliana Nedera, Resident Representative

Steliana Nedera

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I. PURPOSE AND EXPECTED OUTPUT

A. Current situation in Bosnia and Herzegovina with Tuzla Canton specifics

Bosnia and Herzegovina (BiH) is regularly exceeding the guidelines developed by the WHO and the EU's air quality standards, especially in urban areas, and the country has one of the highest mortality rates associated with air pollution in Europe[1]. According to the WB report, in 2019 BiH had the highest air pollution in Europe[2] with GHG emissions per unit of GDP four times higher than EU average[3]. The above has been caused mainly by industrial air emissions and emissions from domestic heating using firewood and lignite.

Residential housing sector in Bosnia and Herzegovina is clearly recognised as the largest final energy consumer. As per latest statistical data, 43% (1.863.7 ktoe) of the overall final energy consumption in B&H in 2017 is attributed to households. Most of it (1.454.9 or 78% of the overall energy consumption) is heat energy and remaining 22% (408,9 ktoe) is electricity.

The SMEs sector in BiH consumes on average up to 35% more energy for their production processes than required (based on conducted energy audits for various projects in the SMEs industry sector in BiH).

Tuzla Canton is **the most populated canton** in FBiH with the population of 445.028 as per B&H Census 2013. At the same time, Tuzla Canton is **the most polluted** part of the country, due to intense and wide-spread use of fossil fuels (coal) for heating in residential and public buildings, and for economic activities run by small, medium and large enterprises, especially those operating in manufacturing field. Severe air pollution in 6-month heating season, causing immense health problems to the population and environmental degradation, is the major environmental issue inhabitants have been facing for decades.

As per B&H Census 2013, total number of occupied residential buildings in Tuzla Canton is 141.947. Out of this number, 138.249 are individual houses and 3.698 multi-dwelling blocks with three and more dwellings.

There are 21.021 businesses (small, medium and large) registered in Tuzla Canton. Out of this number, there are 11.442 crafts (legally recognized as „natural persons") and 9.579 legal entities out of which 98% are micro, small and medium enterprises, while remaining 2% are large enterprises. Most of service-oriented crafts and SMEs in Tuzla Canton are focused on trade, while manufacturing sector is diverse (food-processing, metal-processing and machinery production, wood-processing, textile and clothing, footwear, chemical industry, etc.).

In its Climate Change Adaptation and Low Emission Development Strategy for B&H[4], Nationally Determined Contribution (NDC) under the Paris Agreement[5], and the Third National Communication of B&H under the

[1] http://documents1.worldbank.org/curated/en/117281576515111584/pdf/Air-Quality-Management-in-Bosnia-and-Herzegovina.pdf
[2] idem
[3] Climate Change Adaptation and Low-Emission Development Strategy for Bosnia and Herzegovina, 2013
[4] idem
[5] https://www4.unfccc.int/sites/ndcstaging/PublishedDocuments/BosniaandHerzegovina First/INDC Bosnia and Herzegovina.pdf

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United Nations Framework Convention on Climate Change[6] Bosnia and Herzegovina clearly recognise potential of residential sector for GHG emission reduction, but emphasise that co-financing by the authorities is essential for any substantial progress in residential sector.

Importantly, a robust energy efficiency programme is the only efficient vehicle for substantial reduction of alarming air polution, occuring in B&H towns in winter time, due to extensive use of coal and other fossil fuels in 6-month winter time by dwelling owners, especially those in individual family houses. World Bank' data[7] reveal that 99.6% of B&H' population are exposed to PM2.5 air polution exceeding WHO guideline values. WHO' dana further show that 27% of all deaths in Bosnia and Herzegovina are attributable to the environment[8].

B. Relevant past and current activities of the Tuzla Canton Government

Based on the above statistical facts and below elaborated activities, Tuzla Canton has been selected as the first canton where implementation of decarbonisation measures based on developed financial mechanism will be implemented.

In the past several years, activities related to introduction of energy efficiency in residential buildings such as energy improvements on facade, replacement of windows and doors, replacement of coal-based heating systems with biomass-based heating systems, etc. have been noted. However, due to lack of financial subsidies and incentives, these activities (and their environmental and energy-related benefits) are taking very slow pace. Similar situation is in business sector, with very few enterprises and crafts being able to finance and apply energy efficiency measures.

As a result of continuous public pressure demanding higher air quality and based on a positive experience gained in co-implementation of EU (IPA Interreg)-funded SMART SCHOOLS project, Tuzla Canton Government has launched several co-financing programs: in summer 2020 they launched its first call for co-financing energy efficiency measures in residential sector e.g. for fuel switch; in Sept 2020 they launched a call for co-financing environmental protection measures, including energy efficiency, for SMEs, using funds earmarked for environmental protection. However, their approach to problem solving in energy and environment fields is still based on *ad hoc* decisions, without establishment of strategic long-term approach. Although these initiatives are non-structured and without clear selection criteria, it shows great commitment and initiative of cantonal government.

Tuzla Canton is also taking holistic and planning approach towards environmental protection and decarbonisation through development of strategic documents: with UNDP assistance Local Environmental Action Plans (LEAPS) have been developed in 5 municipalities of Tuzla Canton; Sustainable Energy Action Plan (SEAP) has been developed for the City of Tuzla; and Sustainable Energy and Climate Action Plan (SECAP) also for Tuzla City is in its final stage. Under GCF project *Scaling-up Investment in Low-Carbon Public Buildings*, UNDP is currently supporting preparation of *Energy Efficiency Plan for Energy Renovation of all Public Buildings* under jurisdiction of Tuzla Canton Government. This Plan will enable Tuzla Canton Government to apply for an EBRD loan in the amount of EUR 10 mil.

C. Further expansion of decarbonisation activities

As explained above, Tuzla Canton has shown an initiative to combat climate changes and environmental threats via decarbonisation activities, including energy efficiency measures. So far, project will focus on two Cantons (Tuzla Canton and West Herzegovina Canton) that have expressed a readiness to participate in these activities. Primary focus is on Tuzla Canton, where certain ad hock activities in dealing with air pollution and increase of energy efficiency in residential and SME sector have been implemented. However, a lot of work still needs to be done in terms of establishment of sustainable financial mechanism, stable funding, increase of sets of EE

[6] https://www.undp.org/content/dam/bosnia_and_herzegovina/docs/News/E&E Sector/TNC/TNC Report ENG.pdf

[7] https://data.worldbank.org/country/bosnia-and-herzegovina

[8] WHO definition of „the number of deaths attributed to the environment“This % represents burden of desease that could be avoided by modifying the environment; https://www.who.int/data/gho/indicator-metadata-registry/imr-details/4653

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measures, evidence-based investments, etc. Tuzla Canton has set aside funding for this pilot project, and all activities, from assessment, through establishment and adoption of financial mechanisms to implementation of EE measures in residential and SME sector will be implemented. However, to test replicability potential, willingness of other Cantons and various government administration levels, West Herzegovina Canton was selected where UNDP will initiate same activities in order to establish better baseline of current situation for replication and scale up potential. West Herzegovina Canton is much smaller than Tuzla Canton, just for illustration population of Tuzla Canton is 445.028 while the population of West Herzegovina Canton is 94,000. Due to difference in size, complexity, readiness to co-finance implementation of EE measures and baseline situation more focus will be given to Tuzla Canton while in parallel, same activities and therefore capacity building in West Herzegovina will be implemented. In addition, Federal Ministry of Spatial Planning has shown an intention to co-finance activities in these cantons, but details are yet to be determined. As these two additional cantons, which are under consideration, are smaller in terms of size and complexity than Tuzla canton, this one- or two-month required to select and determine financing arrangements with another canton, will not jeopardize project completion. Pilot projects in both cantons will be completed at the same time. Outputs and results described in this document for Tuzla Canton would be the same for additional canton with the difference in the number of households and SMEs to be supported.

D. UNDP's capability to support decarbonisation process

UNDP has an extensive experience in creating a favourable environment for financing energy efficiency measures which contribute to GHG emission reduction and air quality. In addition to providing technical assistance to institutionalize energy management, improve legislative framework and support development of financial mechanisms (revolving fund, ESCO), UNDP has also vast experience in implementation of infrastructural measures. Only GED programme has resulted in almost 2,000 green jobs created, USD 3 mil saving annually due to increased energy efficiency and 12,500 t of reduced CO_2 emission per year.

In September 2020, within **GED 1 project**, UNDP launched a pilot focusing on improvement of energy efficiency in residential buildings in the cities of Tuzla and Banja Luka. The aim of the project is creation of environment for investment in residential sector in Bosnia and Herzegovina. This pilot projects will result in development of financial mechanisms and guidelines for investments in energy efficiency measures in residential sector at local level. Study on residential stock, establishment of financial mechanisms and guidelines will be developed and further implemented by city administration. During the implementation of this project, GED I will invest app 215,000 USD into development of financial mechanisms and guidelines in Banja Luka and Tuzla cities and USD 200,000 into piloting of these mechanisms. This new initiative on cantonal level is fully in line with pilot project on local level and in fact, it is a scaling up and further strengthening UNDP initiatives in the country.

In line with the above, **GED 3 project** proposal has included activities which focus not only on public sector, but also on residential and SME sectors. If approved, the project will focus on development of financial mechanisms and guidelines on local level (in 12 cities and municipalities which have developed SECAPs and committed to investing funds in residential sector in their cities/ municipalities). This would further strengthen initiative to replicate and scale up project in the country, as these cities/municipalities cover major regions in the country and will serve as positive example for others to join in this initiative.

UNDP has been intensively working with **Sarajevo Canton** on combating air pollution and excessive and non-efficient energy use. Within **GED 2 project**, UNDP is developing a *Study on Improvement of Energy Efficiency in 1,368 Households* used by people in social need. Experiences and lessons learned of the Study will be used in cantons selected for this pilot and will be further strengthened and incorporated in financial mechanism and guidelines, which will include socially vulnerable groups in the mechanisms for support in implementation of energy efficiency in residential sector. UNDP is also developing a *Strategy for Restrictions on the Use of Coal and Other Solid Fuel in Sarajevo Canton for the period 2021-2031*, and a *Model for Subsidizing the Replacement of Coal and Other Solid Fuels Furnaces with Certified Furnaces and Heat Pumps* in households in the same Canton. Total value of these activities is USD 710,000 and their findings and results will be used for further strengthening financial model for Tuzla and one more selected Canton.

E. THE PROPOSED INTERVENTION

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General description and objectives

The intervention aims to support authorities to ensure sustainable implementation of energy efficiency and decarbonization measures. Proposed measures are completely aligned with UNDP's integrated response to COVID 19 and will contribute to COVID "green recovery" by providing technical and economic stimulus to GHG emission reduction, air quality improvement, creation of new jobs, budgetary savings/ re-generating resources for investment etc.

This proposal focuses on development and implementation of mechanisms for financing energy efficiency and decarbonization measures in residential and SME sectors. Techno-economic analysis will serve as a basis for development of this Mechanism and will provide overview of current situation in these two target sectors, proposed energy efficiency measures, cantonal financial and organizational mechanisms for their implementation and their energy-related, environmental (including air quality), fiscal and job-creation benefits.

By implementing structured measures and building system that is well designed we would be able to achieve sustainable and measurable results that could be replicated elsewhere in the country and scaled up on all government levels in Bosnia and Herzegovina.

Scope of the intervention

Output: Authorities are supported to ensure sustainable implementation of energy efficiency and decarbonization measures

Result 1: Mechanisms for financing energy efficiency and decarbonization measures in residential and SME sectors accepted by cantonal authorities (2 cantons)

The following actions will be taken in order to achieve the above result:

- Conduct techno-economic analysis of residential and enterprise sectors in selected cantons, which include baseline inventory of energy consumption and pertaining GHG emissions and air pollution
- Identify and analyze EE improvements for residential and SME sectors (with its energy-related, environmental, fiscal, job-creation benefits)
- Conduct analysis on required financing for implementation of measures (required funds, funding sources, organizational structure, etc.),
- Develop financial mechanism for energy efficiency at Cantonal level; This mechanism will be linked to the existing activities of the Federal Ministry for Spatial Planning, Federal Environmental Fund through its revolving fund and utilization of air emission fees, existing activities of municipalities in selected cantons such as subsidies for fuel switch, and will explore opportunities with national banks and IFI's.
- Develop local guidelines for implementation of EE in residential sector
- Develop local guidelines for implementation of EE in SME sector.

Result 2: Developed financial mechanism that will be successfully piloted on app 100 households and residential buildings

There will be standard energy efficiency measures defined in the mechanism (thermal insulation of facades, replacement of windows and doors, replacement of coal/oil-based heating systems with systems based on renewable energy, etc.).

It is envisaged that some **100 households** and residential buildings (multi-apartment buildings) located in the Tuzla and West Herzegovina Cantons will be selected through a public call for participation. It is anticipated that average total investment per building is 7,000 USD, and that 50% will be project-based provision of financial resources (UNDP 30% and Government 20%) while remaining 50% will be covered by the beneficiaries. These funds will be implemented through the Mechanism(s) developed under Result 1. Due to the fact that some *ad hoc* initiatives have been implemented in Canton Tuzla, and that some experiences and knowledge exist within the Cantonal administration, additional efforts will be made, and capacities raised within Cantonal administration for implementation of energy efficiency in residential and SME sector. Partially these will be achieved through implementation/ development of Financial Mechanisms and through implementation of EE measures. It is envisioned that UNDP will conduct micro assessment (HACT) of relevant Cantonal administration to include responsible ministry as a Responsible Party for project

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activity related to pilots implementation. Responsible Party will be in charge of implementation of EE measures, following previous experience/modality, improved by Financial Mechanism, findings and recommendations of the Study (techno-economic analyses, identified and recommended EE improvements etc.) and established criteria and developed documentation for implementation of pilot projects in residential and SME sectors. So far, these efforts implemented by Cantonal authorities were based on ad hoc measures, financing only one EE measure, concretely replacement of fossil fuel boilers with boilers on renewable energy. Model was designed in a way that Cantonal administration, through public call, made available subsidy funds for implementation of this measure, where final beneficiary was responsible to procure and install, and subsequently justify that the new equipment is functional, which made them eligible for refund of up to 50%. Within this project, as mentioned above, the modality will remain the same, but with significant change shown through established sustainable mechanisms, established clear and transparent criteria, expansion of EE measures that could be funded, established continuous funding etc.

The following actions will be taken in order to achieve the above result:

- Establish criteria for public call for implementation of measures, assuring these criteria are inclusive and widely known to local residents.
- Implementation of tendering procedures for required energy efficiency/ decarbonization works and services.
- Execution of selected projects.
- External technical supervision of executed works.

Result 3: Developed financial mechanism successfully piloted to approximately 15 SMEs

As a result of Result 1, conducted techno-economic analysis of enterprise sector, and Result 2, developed financial mechanisms, as well as EE Action Plan 2016-2018 (and 2019-2021) for Bosnia and Herzegovina, energy efficiency measures will be identified and implemented in enterprise and commercial service sectors. They may include:

- ***Increase of energy efficiency of industrial processes***, which primarily relates to reduction of consumption of electrical and/or heat energy (i.e. replacement of current outdated equipment /machinery and energy-supply systems with energy efficient ones; reconstruction of steam-generating plants, reconstruction of equipment for compressed-air supply, energy-related optimization and automatization of industrial processes, and many others);
- ***Application of co-generation and three-generation in industrial processes***, where co(three)-generation means production at the same time of two useful energies from one energy source (i.e. combined production in one facility of electrical and heat energy);
- ***Own production of energy from renewable energy sources***, established within the SME /craft (i.e. replacement of current fossil-fueled boilers with renewable-energy boilers, installation of solar thermal system for water heating, installation of solar photovoltaic systems for electricity generation, installation of thermal pumps, etc.);
- ***Energy efficiency improvements in buildings owned by target SMEs and crafts***

  While energy efficiency measures in residential sector are quite uniformed, measures in SME sector will be tailored to the specific needs of each business and to specifics of their manufacturing /service activities

It is envisaged that some **15 SME's** and crafts located in the Tuzla and West Herzegovina Cantons will be selected through a public call for participation. It is anticipated that average total investment per business is 20,000 USD, and that 60% would be project-based provision of financial resources (provided by UNDP 40% and government 20%) while remaining 40% will be covered by the beneficiaries. These funds will be also implemented through the Mechanism(s) defined under Result 1 and within the same modality as described under Result 2.

The following actions will be taken in order to achieve the above result:

- Establish criteria for public call for implementation of measures, assuring these criteria are widely known to local SME's operators.

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- Implementation of tendering procedures for required energy efficiency/ decarbonization works and services.
- Execution of selected projects.
- External technical supervision of executed works.

Result 4: Results of pilot project are documented at presented to relevant audience

Institutional knowledge gained in the pilot implementation in residential and SME sectors**,** with emphasis on those enabling wide-scale replicability in other FBiH cantons and Republika Srpska will be elaborated and documented. These analyses will focus on pilot results such as savings/re-generating resources for investment, enabling partnerships with banks and other financing institutions and their replicability potential to achieve results at scale.

The following actions will be taken in order to achieve the above result:

- Analysis of energy-related and financial outcomes of the implemented projects in residential sector
- Analysis of energy-related and financial outcomes of the implemented projects in SME sector
- Presentation of results.

II. MANAGEMENT ARRANGEMENTS

Duration: 01 Nov 2020 – 31 Oct 2021

Initiation Plan duration: 12 months

Project management

The project will be implemented following UNDP's Direct Implementation Modality **(DIM)**. The **Implementing Partner** for this project is *UNDP.* The Implementing Partner is responsible and accountable for managing this project, including the monitoring and evaluation of project interventions, achieving project outcomes, and for the effective use of UNDP resources. UNDP will ensure that all partners and subcontractors will be selected based on open and transparent selection processes, ensuring: i) a clear link between implementation and policy components, ii) cost-effectiveness, iii) the sustainability of capacity building measures.

The **Project Board** is responsible for making by consensus, management decisions when guidance is required by the Project Manager, including recommendation for UNDP/Implementing Partner approval of project plans and revisions. In order to ensure UNDP's ultimate accountability, Project Board decisions should be made in accordance with standards that shall ensure management for development results, best value for money, fairness, integrity, transparency and effective international competition. In case a consensus cannot be reached within the Board, final decision shall rest with the UNDP Programme Manager. The Project Board is comprised of the following institutions: Representatives of Canton Tuzla, Ministry of Fereign Trade and Economic Relations and UNDP.

Project team structure

The **Project Team** will be led by the **Project Manager** (100 % work-time), who will be responsible for overall Project coordination and day-to-day management of project activities and will ensure that the Project produces the results specified, to the required corporate standards and within the constraints of time and cost.

The **Project Manager** will run the project on a day-to-day basis on behalf of the Implementing Partner within the constraints laid down by the Board. The Project Manager function will end when the final project terminal evaluation report and other documentation required by UNDP, has been completed and submitted to UNDP (including operational closure of the project).

The **Project Assurance** role will be provided by *UNDP Country office Energy and Environment Sector Leader.*

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The overall financial, administrative and logistical support will be ensured via **1 Project Assistant** (UNDP Office, 100 % worktime). Regular communication and interaction will be ensured between the Project team, particularly among the staff based within the UNDP and the BIH stakeholders.

In addition, the Project will deploy key short-term national and international specialists to support Project implementation. Key experts required for implementation of the project are as follows:

- **Civil Engineer – National expert for development of residential and SMEs study**

National expert with minimum of 5 years of relevant professional experience in the fields of energy efficiency, climate change, building reconstruction management & implementation, and strategic planning at local and national levels, preparation of EE studies and similar strategic documents in the field of energy efficiency and renewable energy sources.

The Studies must provide crucial information and entry-data to the Cantonal authorities, required for creation of long-term financial & policy mechanisms for large-scale investment in low-carbon residential buildings and SME sector in selected Cantons. Therefore, the Civil Engineer will obtain information on building characteristics for each analysed building, and for each unit within the analysed building (floor area, types of material used for making the building, total surface of heated part of building, information on current consumption, etc.). Furthermore, Civil Engineer will use analysis to prioritize buildings and propose measures to improve energy efficiency.

- **Mechanical Engineer – National expert for development of residential and SMEs study**

National expert with minimum of 5 years of relevant professional experience in the fields of energy efficiency, climate change, building reconstruction management & implementation, and strategic planning at local and national levels, preparation of EE studies and similar strategic documents in the field of energy efficiency and renewable energy sources.

The Studies must provide crucial information and entry-data to the City Authorities in Banja Luka and Tuzla, required for creation of long-term financial & policy mechanisms for large-scale investment in low-carbon residential buildings in these two cities. Therefore, the Mechanical Engineer will work on obtaining information on building characteristics for each analysed building, and for each unit within the analysed building (floor area, types of material used for making the building, total surface of heated part of building, information on current consumption, etc.). Furthermore, Mechanical Engineer will use analysis from mechanical engineering perspective to prioritize buildings and propose measures to improve energy efficiency. The Mechanical Engineer will also contribute to the development of financial mechanisms, in accordance with results of the Studies.

- **Electrical Engineer – National expert for development of residential and SMEs study**

National expert with minimum of 5 years of relevant professional experience in the fields of energy efficiency, climate change, building reconstruction management & implementation, and strategic planning at local and national levels, preparation of EE studies and similar strategic documents in the field of energy efficiency and renewable energy sources.

The Studies must provide crucial information and entry-data to the City Authorities in Banja Luka and Tuzla, required for creation of long-term financial & policy mechanisms for large-scale investment in low-carbon residential buildings in these two cities. Therefore, the Civil Engineer will work on obtaining information on building characteristics for each analysed building, and for each unit within the analysed building (floor area, types of material used for making the building, total surface of heated part of building, information on current consumption, etc.). Furthermore, Electrical Engineer will use analysis from electrical engineering perspective to prioritize buildings and propose measures to improve energy efficiency. The Electrical Engineer will also contribute to the development of financial mechanisms, in accordance with results of the Studies.

- **Economist – National expert for development of residential and SMEs study**

National expert with minimum of 5 years of relevant professional experience in the fields of energy efficiency, climate change, building reconstruction management & implementation, and strategic planning at local and national levels, preparation of EE studies and similar strategic documents in the field of energy efficiency and renewable energy sources.

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Each Study shall serve as a reliable planning tool, enabling the City authorities to get clear insight in entire residential stock on their territory and scope of pertaining energy-saving and emission-reduction opportunities, and to identify, based on the economic evaluation of investment and effects of proposed energy efficiency measures, priority buildings and measures to improve energy efficiency. Therefore, the Economist will need to work in cooperation with all Engineers to analyse the amount of investment required for implementation of proposed energy efficiency measures and consequently develop economic evaluation indicators of the proposed energy efficiency measures for each analysed building, such as simple payback period, net present value, internal rate of return, profitability index, and their energy &economy-related parameters. The Economist will develop new subsidy mechanisms/model for low-income households, including an analysis of the existing model and new subsidy approach after implementation of the proposed EE measures. He/she will have a primary role in development of financial mechanisms, in accordance with results of the Studies.

- **Legal Expert - National expert for development of residential and SMEs study**

National expert with minimum 5 years of professional experience in in development of legal and policy documents with proven experience in area of Energy & Environment and/or EU Projects.

The Legal Expert will provide comprehensive legal support in the designing and developing of financial mechanisms. Legal Expert will conduct in-depth research based on a desk review of available literature and analyse legislative frameworks, which can directly affect development/implementation of financial mechanisms in BiH. He/she will also ensure financial mechanisms are constructed within existing legal boundaries and provide effective legal solutions in accordance with BiH legislation. The financial mechanisms must be harmonized based on key outcomes from the two Studies and required energy-saving targets planned. Consequently, the Legal Expert will provide legal contribution in the development of sustainable and long-term financial mechanisms for improving energy efficiency in the residential housing sector.

III. MONITORING

Initiation Phase Report should be prepared at the end of the Initiation Plan, using the standard format available in the Executive Snapshot in addition to attaching the full project document if developed or an explanation if initiation failed to produce a full project.

In addition, the initiative will be monitored based on specific indicators for each Activity.

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IV. WORK PLAN

Period: November 01, 2020 to October 31, 2021 (12 months)

Goal: in collaboration with UNDP, authorities are supported to ensure sustainable implementation of energy efficiency and decarbonization measures.

EXPECTED OUTPUTS And baseline, indicators including annual targets	PLANNED ACTIVITIES List activity results and associated actions	TIMEFRAME				RESPONSIBLE PARTY	PLANNED BUDGET		
		Q1	Q2	Q3	Q4		Funding Source	Budget Description	Amount

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Output: Authorities are supported to ensure sustainable implementation of energy efficiency and decarbonization measures ***Baseline:*** Extremely high level od air pollution, *n*o systemic and sustainable approach towards energy efficiency and decarbonisation in Bosnia and Herzegovina ***Indicators:*** Extent to which authorities in Bosnia and Herzegovina support sustainable implementation of energy efficiency and decarbonization measures Number of households benefitting from implemented energy efficiency/ decarbonization works and services, out of which at least 30% are females. Number of companies benefitting from implemented energy efficiency/ decarbonization works and services. ***Targets:*** Financial mechanism for energy	**Activity 1.** Mechanisms for financing energy efficiency and decarbonization measures in residential and SME sectors accepted by cantonal authorities (2 cantons) - Conduct techno-economic analysis of residential and enterprise sectors in selected cantons, which include baseline inventory of energy consumption and pertaining GHG emissions and air pollution - Identify and analyze EE improvements for residential and SME sectors (with its energy-related, environmental, fiscal, job-creation benefits) - Conduct analysis on required financing for implementation of measures (required funds, funding sources, organizational structure, etc.), - Develop financial mechanism for energy efficiency at Cantonal level; This mechanism will be linked to the existing activities of the Federal Ministry for Spatial Planning, Federal Environmental Fund through its revolving fund and utilization of air emission fees, existing activities of municipalities in selected cantons such as subsidies for fuel switch, and will explore opportunities with national banks and IFI's. - Develop local guidelines for implementation of EE in residential sector - Develop local guidelines for implementation of EE in SME sector.					UNDP	UNDP	Contractual services-individuals Contractual service - companies	1,035,000

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efficiency developed at Cantonal level At least 100 households benefitting from implemented energy efficiency/ decarbonization works and services. At least 15 companies benefitting from implemented energy efficiency/ decarbonization works and services. Related CP outcome: By 2020, legal and strategic frameworks are enhanced and operationalized to ensure sustainable management of natural, cultural and energy resources international obligations and enforced at entity and state levels	**Activity 2.** Developed financial mechanism successfully piloted on app 100 households and residential buildings- Establish criteria for public call for implementation of measures, assuring these criteria are inclusive and widely known to local residents.- Implementation of tendering procedures for required energy efficiency/ decarbonization works and services.- Execution of selected projects.- External technical supervision of executed works.					UNDP, cantonal government	UNDPGovernment of BiH (multiple levels)Private sector	Contractual services-individuals Contractual service – companies Equipment	900,000
	Activity 3. Developed financial mechanism successfully piloted on app 15 SME's - Establish criteria for public call for implementation of measures, assuring these criteria are widely known to local SME's operators. - Implementation of tendering procedures for required energy efficiency/ decarbonization works and services. - Execution of selected projects. - External technical supervision of executed works.					UNDP, cantonal government	UNDP Government of BiH (multiple levels) Private sector	Contractual services-individuals Contractual service – companies Equipment	300,000
	Activity 4. Results of pilot project are documented at presented to relevant audience - Analysis of energy-related and financial outcomes of the implemented projects in residential sector - Analysis of energy-related and financial outcomes of the implemented projects in SME sector - Presentation of results					UNDP	UNDP	Contractual services-individuals Workshop	25,000
Total									**2,260,000**

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